Quantification of Trace Metals and Volatile Organic Compounds (VOCs) in Commercially Available Water-Based Paint

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Introduction

Why study paint?
Water-based latex paint is the preferred choice for interior wall applications among homeowners.

Verification of the manufacturer's claim with respect to volatile compounds and **green guard certification**.

Purpose?
The aim of this research project assess, through quantification, the **possible environmental and health risks** associated with commercially available water-based paint. Trace metals will be analyzed using Inductively Coupled Plasma (ICP) instruments and VOCs analyzed using a Gas Chromatography (GC) instrument. Additionally, for metal analysis, we are proposing a "modified" alternative for paint digestion as research provides methods involving digestion using very corrosive acids (HF).

Analytes?
Trace metals are chemical elements available in a very small amounts. Heavy metals are subsets of trace metals. They have a have a higher density (ρ) and molecular weight (M_r) relative to water. At increased concentrations, these can become harmful.
→ Cr, Cd, Pb, Cu, K, Mg, Mn, Zn
VOCs are organic compounds with very high vapor pressures at Rt, increasing the rate of volatility. Molecules with BPs >250°C are not considered VOCs. Additionally, they can be carcinogenic and harmful to the envi

Sampling & Sample Preparation

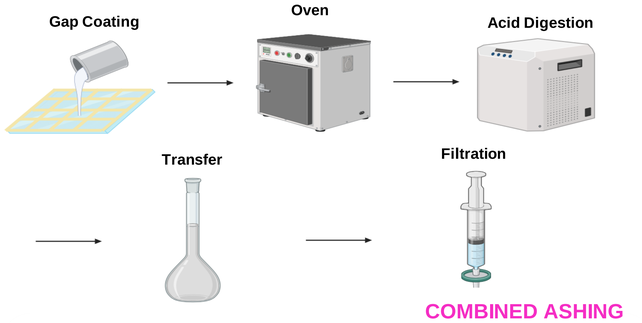
SAMPLES



Figure 1. Sample 1 (Marquee) and Sample 2 (Dynasty)

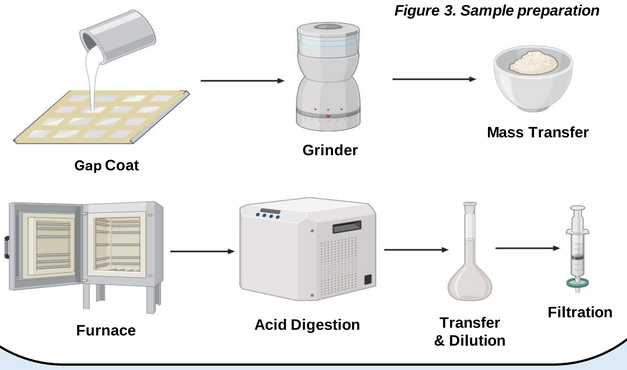
WET ASHING

Figure 2. Sample preparation



COMBINED ASHING

Figure 3. Sample preparation



Data

ICP, SEM + EDX & GC Results

Detection?

Screening on the ICP-OES, at three different wavelengths, for potential analytes using the samples prepared as shown in ***Figure 1*** indicated...

Cr, K, Mn, Pb, Zn, Mn were present in all samples; K and Zn found in **greater concentrations** in sample 'S1'.

As expected however, all analytes were found in trace amounts.

Calibration?

Standards were prepared within a range of **1 - 10,000 ppb** for Cr, Cd, and Pb while the remaining analytes, Cu, K, Mg, Mn and Zn were prepared within a range of **1 - 8,000 ppb**.

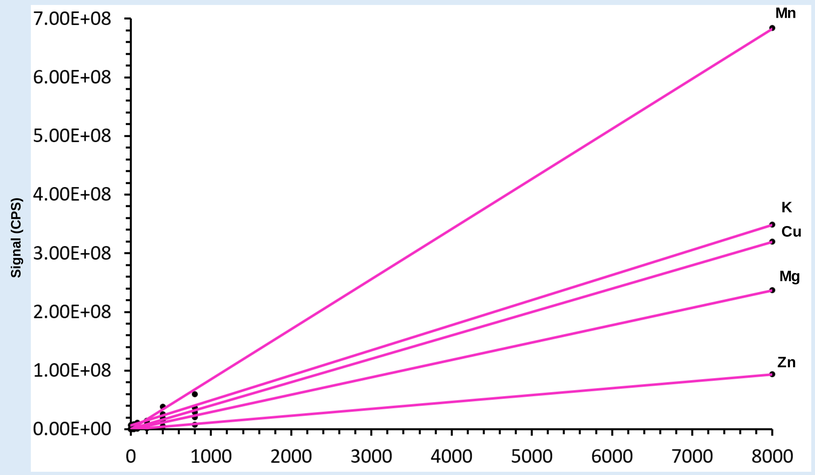


Figure 4. Calibration plot of analytes measured at a 1 – 8,000 ppb range [Cu, Fe, K, Mg, Mn, Zn]

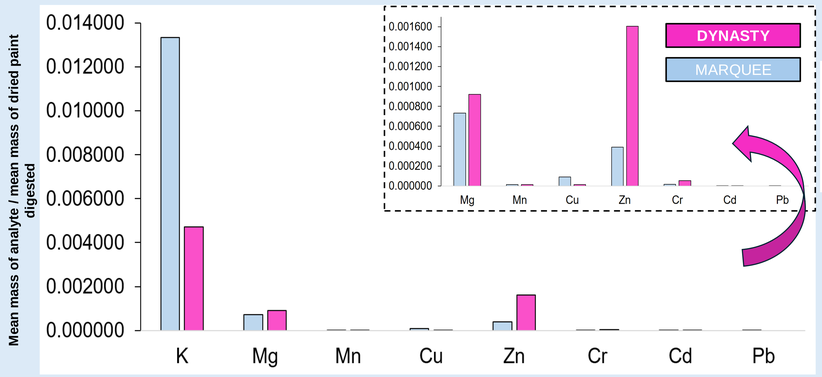


Figure 5. Comparison of measured analytes in BEHR paint samples per mass digested sample

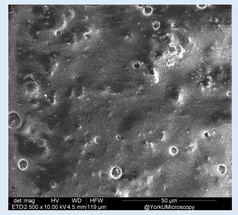


Figure 6. SEM image of the latex film layer of sample S1A

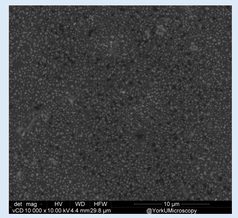


Figure 7. EDX image of the elemental analysis from sample S1B showing mostly Ti (TiO_2)

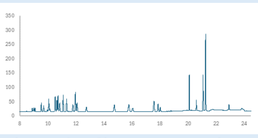


Figure 8. Chromatogram of EPA 524/502.2 Standard on Headspace GC-FID of Identifiable VOCs

Six VOCs were found in tinted Sample 1 whereas **three** VOCs were found in un-tinted Sample 1. (Shared with tinted Marquee)

Additionally, there were no identifiable VOCs found in Sample 2 based on the EPA Standard used..

Table 1. Statistical comparison between measured analytes of BEHR paint samples MARQUEE and DYNASTY

Analyte	K	Mg	Mn	Cu	Zn	Cr	Cd	Pb
R^2	0.9997	0.9997	0.9997	0.9998	0.9997	1.0000	1.0000	0.9553
Equation	y = 42785x + 6E+06	y = 29660x - 139556	y = 85455x - 591533	y = 39898x + 481641	y = 11735x - 59844	y = 60003x - 22398	y = 26382x - 59752	y = 54998x + 773619
RSD (%)	68	16	6	107	86	69	3	146

Discussion

Sampling
The sampling procedure has indicated that ASTM 3335-85a, which highlights a method for digesting paint for trace metal (Pb) analysis, incorporates a more effective method than simply dry ashing. Although this standardized method was not closely followed, the method described in ***Figure 3*** is a modified version of this ASTM, with respect to equipment available within the laboratory. ASTM 6886-2 was unsuccessful in isolating VOC for direct inject.

Data
Ti was not analyzed because approximately 70 % of paint consists of TiO_2, which is an inorganic pigment molecule.

The results have confirmed that K and Zn are two metals that are in a greater concentration in the samples.

From a health and safety perspective, the results indicate that the **concentration of heavy metals** for both paint formulations are **very low**.

Calibration
A calibration plot explain how related the x and y variables relate with one another. For all analytes, these relationships displayed an R^2 greater than 0.9995, expect Pb which was 0.9553.

Future Work

Expand the analytical scope to include a broader range of heavy metals and VOCs relevant to environmental regulations or industry standards.

Thoroughly validate analytical methods for heavy metal and VOC analysis to ensure accuracy, precision, and reliability.

Better separation of VOCs with mid polarity column (e.g DB-624) and identification via mass spectrophotometry.

Increase sample size and replicate analysis to enhance statistical robustness and improve understanding of sample variability.

Implement additional quality control measures, including analysis of CRMs, blanks, and duplicates, to monitor for contamination and ensure method precision.

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